

ANOVA 2

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Topics for this lecture

- ANOVA with several factors
- The factorial applet
- Effect plots
- Estimation and hypothesis testing
- Tukey test and confidence intervals
- ANCOVA - [Example:Regression](#) with indicator variables

ANOVA with several factors

- Each with two or more levels
- Assume two factors: A with $i = 1, 2, \dots, a$ levels, and B with $j = 1, 2, \dots, b$ levels
- If all levels of A are tested with all levels of B we have a complete crossed design.
- Assume there are $k = 1, 2, \dots, n$ replicates of all ab level combinations.

Illustration of two-factor case

<https://solve.shinyapps.io/Factorialdemo/>

The Model

Model:

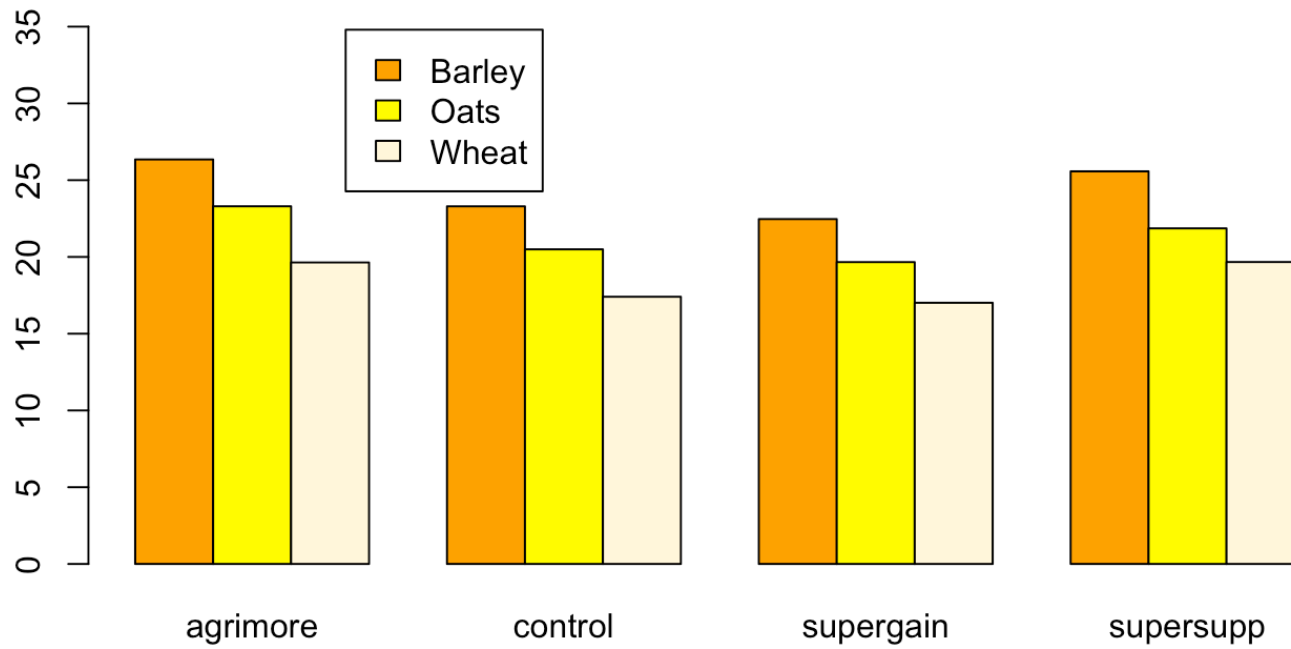
$$y_{ijk} = \mu + \alpha_i + \beta_j + \alpha\beta_{ij} + \epsilon_{ijk}$$

where $\epsilon_{ijk} \sim N(0, \sigma^2)$.

Parametrization (restrictions)

- Sum-to-zero: $\sum_i \alpha_i = 0$, $\sum_j \beta_j = 0$, $\sum_i \alpha\beta_{ij} = 0$ and $\sum_j \alpha\beta_{ij} = 0$.
- Reference levels: $\alpha_1 = 0$, $\beta_1 = 0$, $\alpha\beta_{1,j} = 0$ for all j and $\alpha\beta_{i,1} = 0$ for all i .

Example: Animal diets - The “growth data”



R analysis

For reference level analysis, see “the R-book” on page 517.

Let’s again use the sum-to-zero parametrization here:

```
mod2 <- lm(gain ~ diet + supplement + diet:supplement,  
           data=weights, contrasts = "contr.sum")
```

```
## Warning in model.matrix.default(mt, mf, contrasts): non-list contrasts argument  
## ignored
```

Equivalent model calls in R:

- `gain ~ diet + supplement + diet:supplement`
- `gain ~ diet*supplement`
- `gain ~ (diet + supplement)^2`

Diet means and sd's

```
by(weights$gain, weights$diet,  
    FUN=function(x){c(mean=mean(x),sd=sd(x))})
```

```
## weights$diet: barley
```

```
##      mean      sd
```

```
## 24.421638  2.268806
```

```
## -----
```

```
## weights$diet: oats
```

```
##      mean      sd
```

```
## 21.328822  1.665849
```

```
## -----
```

```
## weights$diet: wheat
```

```
##      mean      sd
```

```
## 18.43134  1.59873
```

Supplement means and sd's

```
by(weights$gain, weights$supplement,  
    FUN=function(x){c(mean=mean(x),sd=sd(x))})
```

```
## weights$supplement: agrimore
```

```
##      mean      sd
```

```
## 23.095311  3.176166
```

```
## -----
```

```
## weights$supplement: control
```

```
##      mean      sd
```

```
## 20.39861  2.71380
```

```
## -----
```

```
## weights$supplement: supergain
```

```
##      mean      sd
```

```
## 19.713853  2.539195
```

```
## -----
```

```
## weights$supplement: supersupp
```

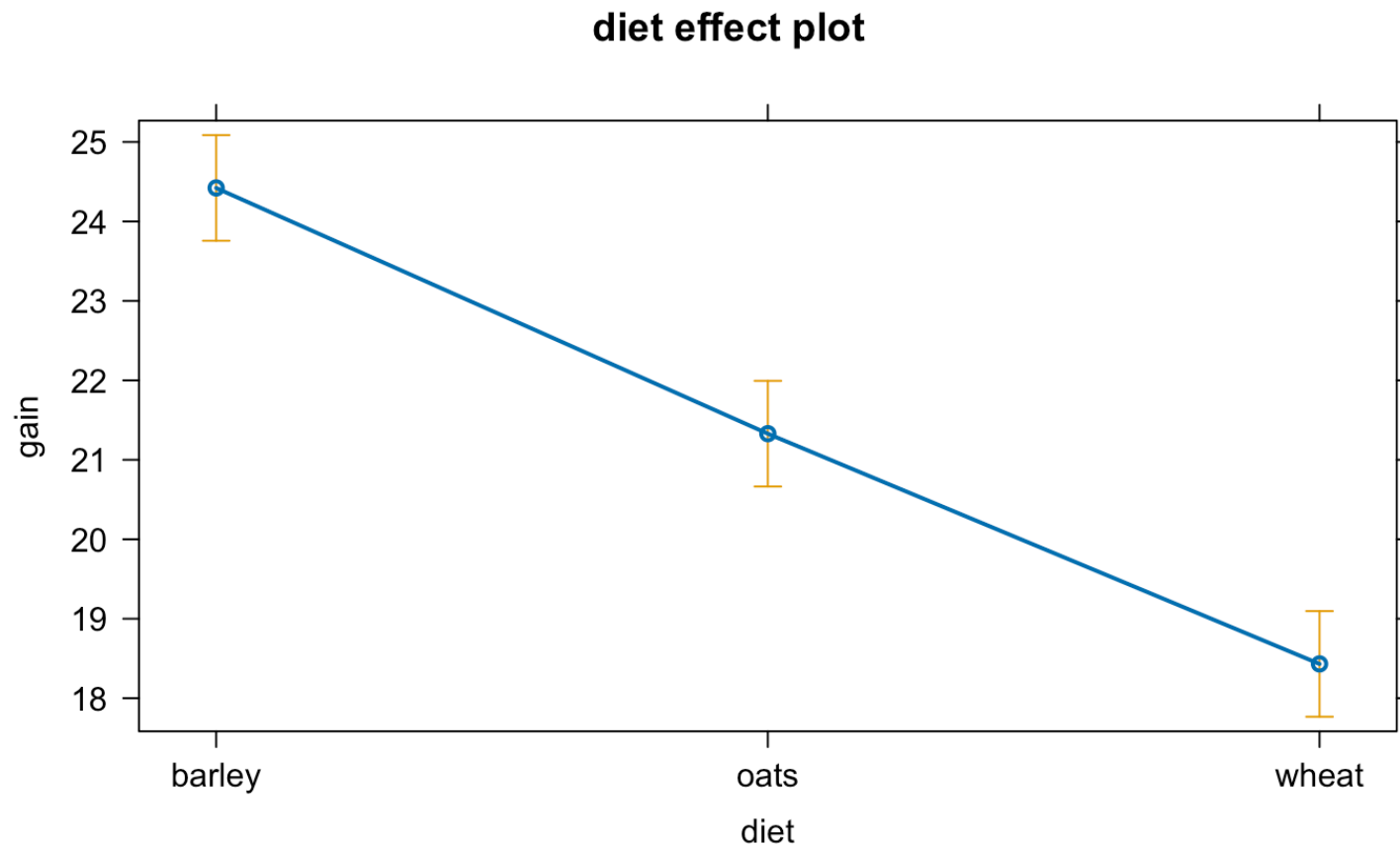
```
##      mean      sd
```

```
## 22.367959  2.853456
```

Main effects plot of **Diet**

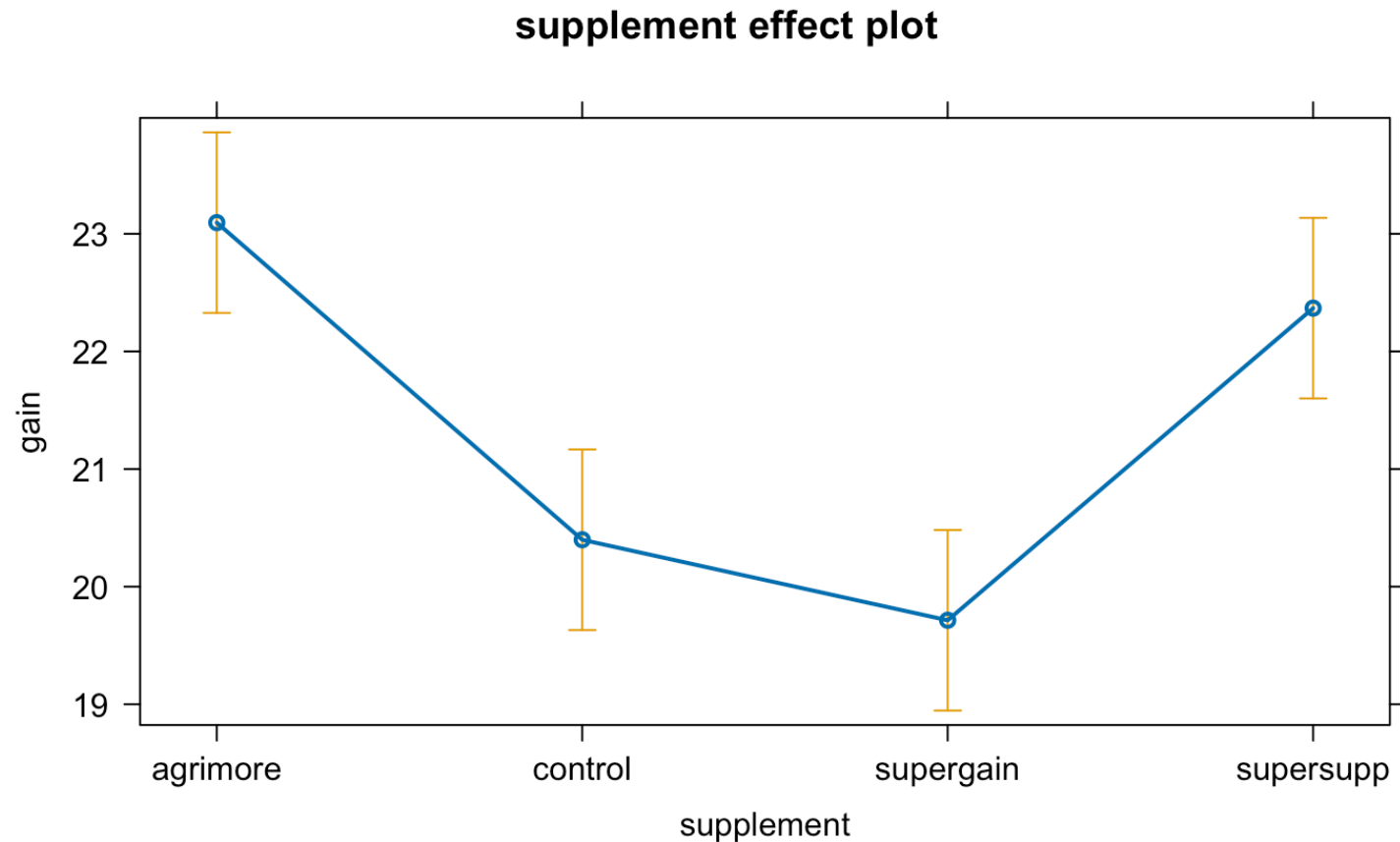
“Main effect plots” of group means with 95% CI's

```
library(effects)  
plot(Effect(focal.predictors = c("diet"), mod = mod2))
```



Main effects plot of Supplement

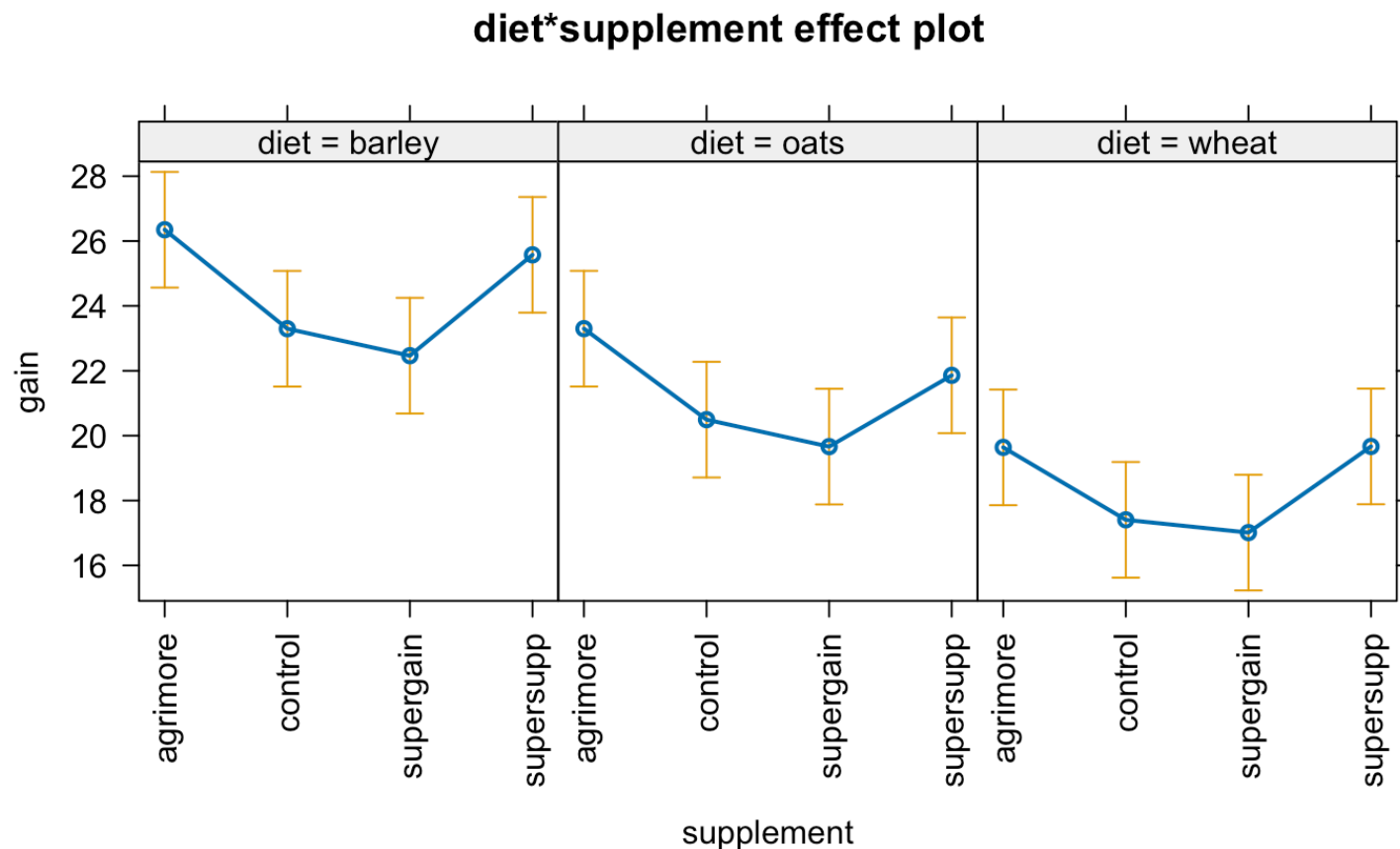
```
plot(Effect(c("supplement"), mod2))
```



Interaction plots

Means of diet:supplement combinations.

```
library(effects)
plot(allEffects(mod2, confidence.level=.99), layout=c(3,1),
     rotx=90, alternating=FALSE)
```



Model parameter estimation

For balanced data the least squares estimators for the model parameters are (sum-to-zero):

$$\hat{\mu} = \bar{y}_{...}$$

$$\hat{\alpha}_i = \bar{y}_{i..} - \bar{y}_{...}$$

$$\hat{\beta}_j = \bar{y}_{.j.} - \bar{y}_{...}$$

$$\hat{\alpha\beta}_{ij} = \bar{y}_{ij.} - \bar{y}_{i..} - \bar{y}_{.j.} + \bar{y}_{...}$$

And as usual, the noise variance estimate is:

$$\hat{\sigma}^2 = MSE$$

Summary with estimates of effects:

##	Estimate	Std. Error
## (Intercept)	26.3484777	0.6555863
## dietoats	-3.0500939	0.9271390
## dietwheat	-6.7094047	0.9271390
## supplementcontrol	-3.0518277	0.9271390
## supplementsupergain	-3.8823540	0.9271390
## supplementsupersupp	-0.7731751	0.9271390
## dietoats:supplementcontrol	0.2471088	1.3111726
## dietwheat:supplementcontrol	0.8182729	1.3111726
## dietoats:supplementsupergain	0.2469737	1.3111726
## dietwheat:supplementsupergain	1.2557124	1.3111726
## dietoats:supplementsupersupp	-0.6649742	1.3111726
## dietwheat:supplementsupersupp	0.8024430	1.3111726

We want to test hypotheses

Interaction effects?

$$H_0 : \alpha\beta_{11} = \alpha\beta_{12} = \cdots = \alpha\beta_{ab} = 0$$

Main effect of A?

$$H_0 : \alpha_1 = \alpha_2 = \cdots = \alpha_a = 0$$

Main effect of B?

$$H_0 : \beta_1 = \beta_2 = \cdots = \beta_b = 0$$

Splitting of sums of squares

The Total Sums of Squares measures total variability:

$$SST = \sum_i \sum_j \sum_k (y_{ijk} - \bar{y}_{...})^2$$

Can be split into:

$$SST = SSA + SSB + SSAB + SSE$$

Degrees of freedom follow correspondingly as

$$(N - 1) = [(a - 1)] + [(b - 1)] + [(a - 1)(b - 1)] + [ab(n - 1)]$$

F-tests

Interaction: Reject H_0 if

$$F = \frac{MSAB}{MSE} = \frac{SSAB/(a-1)(b-1)}{SSE/ab(n-1)}$$

is larger than $F_{\alpha, (a-1)(b-1), ab(n-1)}$ or p-value $< \alpha$.

Main effect A: Reject H_0 if

$$F = \frac{MSA}{MSE} > F_{\alpha, (a-1), ab(n-1)}$$

Main effect B: Reject H_0 if

$$F = \frac{MSB}{MSE} > F_{\alpha, (b-1), ab(n-1)}$$

The ANOVA-table for the growth data

```
anova(mod2)
```

```
## Analysis of Variance Table
```

```
##
```

```
## Response: gain
```

##		Df	Sum Sq	Mean Sq	F value	Pr(>F)	
##	diet	2	287.171	143.586	83.5201	2.999e-14	***
##	supplement	3	91.881	30.627	17.8150	2.952e-07	***
##	diet:supplement	6	3.406	0.568	0.3302	0.9166	
##	Residuals	36	61.890	1.719			

```
## ----
```

```
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Conclusions?

A reduced model

```
mod3 <- lm(gain ~ diet + supplement, data=weights, contrasts = "contr.sum")

## Warning in model.matrix.default(mt, mf, contrasts): non-list contrasts argument
## ignored

anova(mod3)

## Analysis of Variance Table
##
## Response: gain
##           Df  Sum Sq Mean Sq F value    Pr(>F)
## diet        2 287.171  143.586   92.358 4.199e-16 ***
## supplement   3  91.881   30.627   19.700 3.980e-08 ***
## Residuals  42   65.296    1.555
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Note: The complete crossed design gives “orthogonal effects”. The SS’s are independent, and, apart from SSE, they do not change when effects are removed/added to the model.

Tukey tests supplement - Confidence intervals

```
##           Estimate      lwr      upr
## agrimore-control  2.6967005  1.3350677  4.0583332
## agrimore-supergain 3.3814586  2.0198259  4.7430914
## agrimore-supersupp 0.7273521 -0.6342806  2.0889849
## control-supergain  0.6847581 -0.6768746  2.0463909
## control-supersupp -1.9693484 -3.3309811 -0.6077156
## supergain-supersupp -2.6541065 -4.0157392 -1.2924737
## attr(,"conf.level")
## [1] 0.95
```

Compact letter display

A compact letter display (cld) of groups of effects

```
cld(ptests)
```

```
## Tukey's HSD
```

```
## Alpha: 0.05
```

```
##
```

```
##           Mean G1 G2
```

```
## agrimore  23.09531    B
```

```
## supersupp 22.36796    B
```

```
## control   20.39861    A
```

```
## supergain 19.71385    A
```

Analysis of covariance - ANCOVA

The dataset `birth` in the `BIAS.data` package holds records of 189 birth weights from Massachusetts, USA, and some additional variables. Three of the variables are,

Var	Description
LWT	Weight of mother.
SMK	YES if mother smokes and NO otherwise.
BWT	Birth weight (RESPONSE).

The main question is: Does mother's smoking affect birth weight and does it depend on the mother weight?

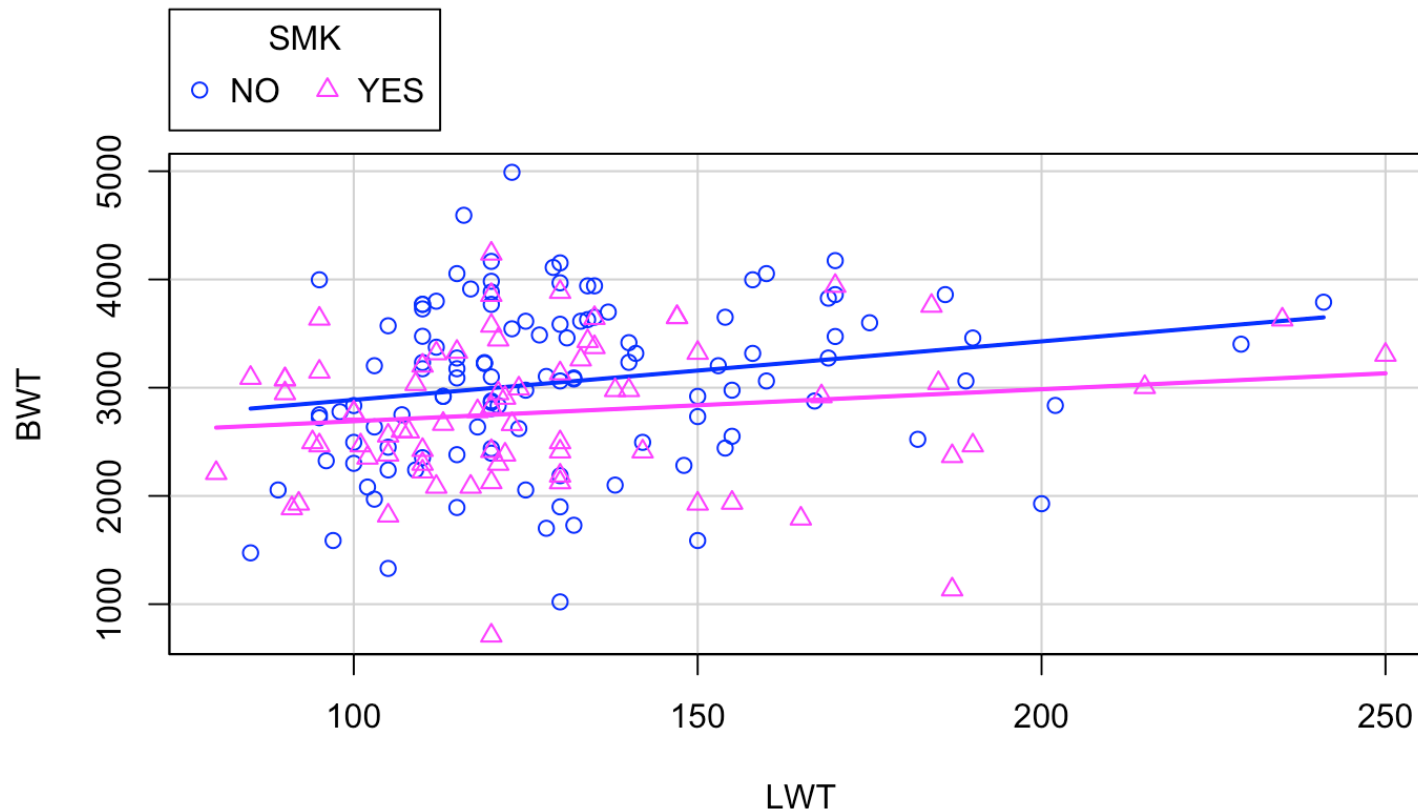
We have both a categorical and a continuous predictor for the response variable.

The data

##		LWT	SMK	BWT
##	1	120	YES	709
##	2	130	NO	1021
##	3	187	YES	1135
##	4	105	NO	1330
##	5	85	NO	1474
##	6	150	NO	1588
##	7	97	NO	1588

Scatterplot with separate linear models for smokers and non-smokers

```
## Warning in applyDefaults(regLine, defaults = list(method = lm, lty = 1, :  
## unnamed arguments, will be ignored
```



ANCOVA as regression with indicator variable

Let $y = BW T$ and $x_1 = LWT$ and define

$$x_2 = \begin{cases} 0 & \text{if smoker} \\ 1 & \text{if non-smoker} \end{cases}$$

Assume the model

$$y_i = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + \epsilon_i$$

where $\epsilon_i \sim N(0, \sigma^2)$.

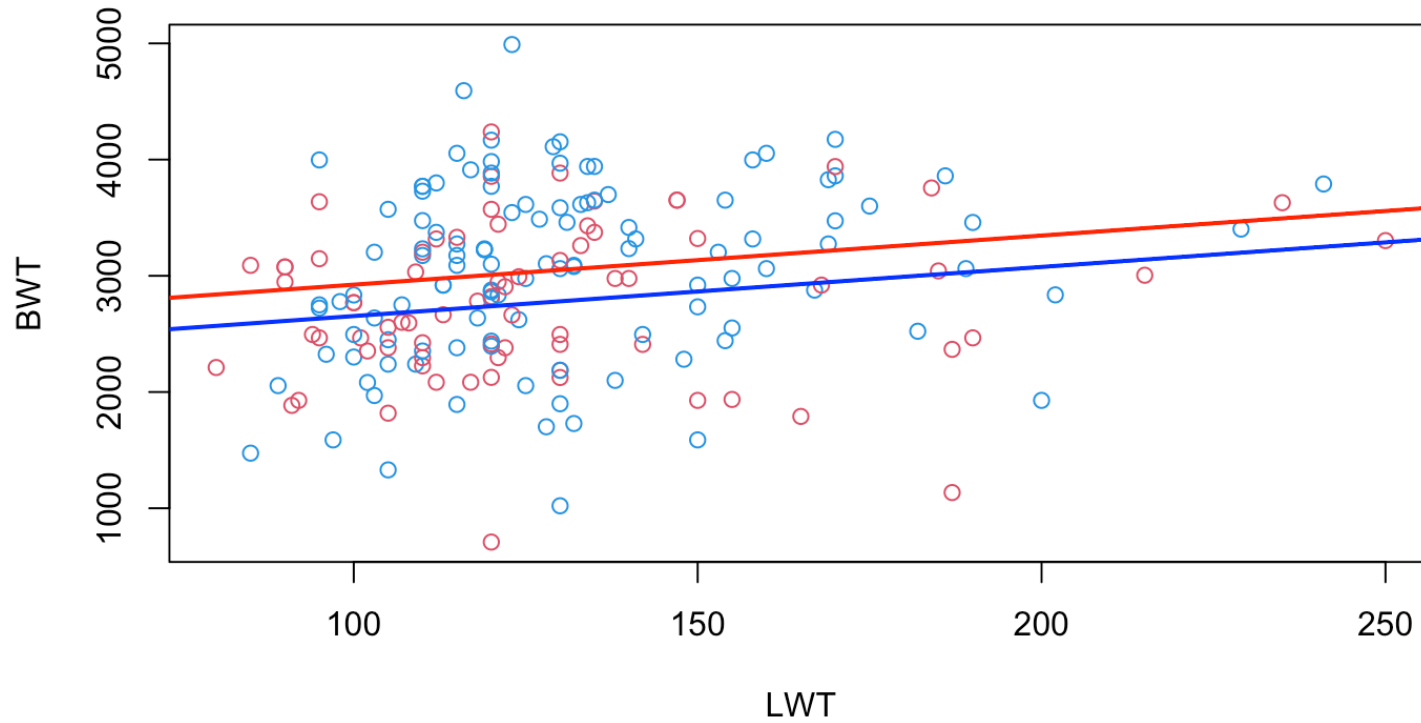
Fitting the model in R

```
birth4 <- lm(BWT ~ LWT + SMK, data=birth, contrasts = "contr.treatment")
birth4
```

```
##
## Call:
## lm(formula = BWT ~ LWT + SMK, data = birth, contrasts = "contr.treatment")
##
## Coefficients:
## (Intercept)          LWT          SMK(YES)
##   2500.174         4.238        -270.013
```

Explain coefficients

The estimated model(s)



The model with interaction term

Assume that we expand the model with an interaction term between LWT and SMK

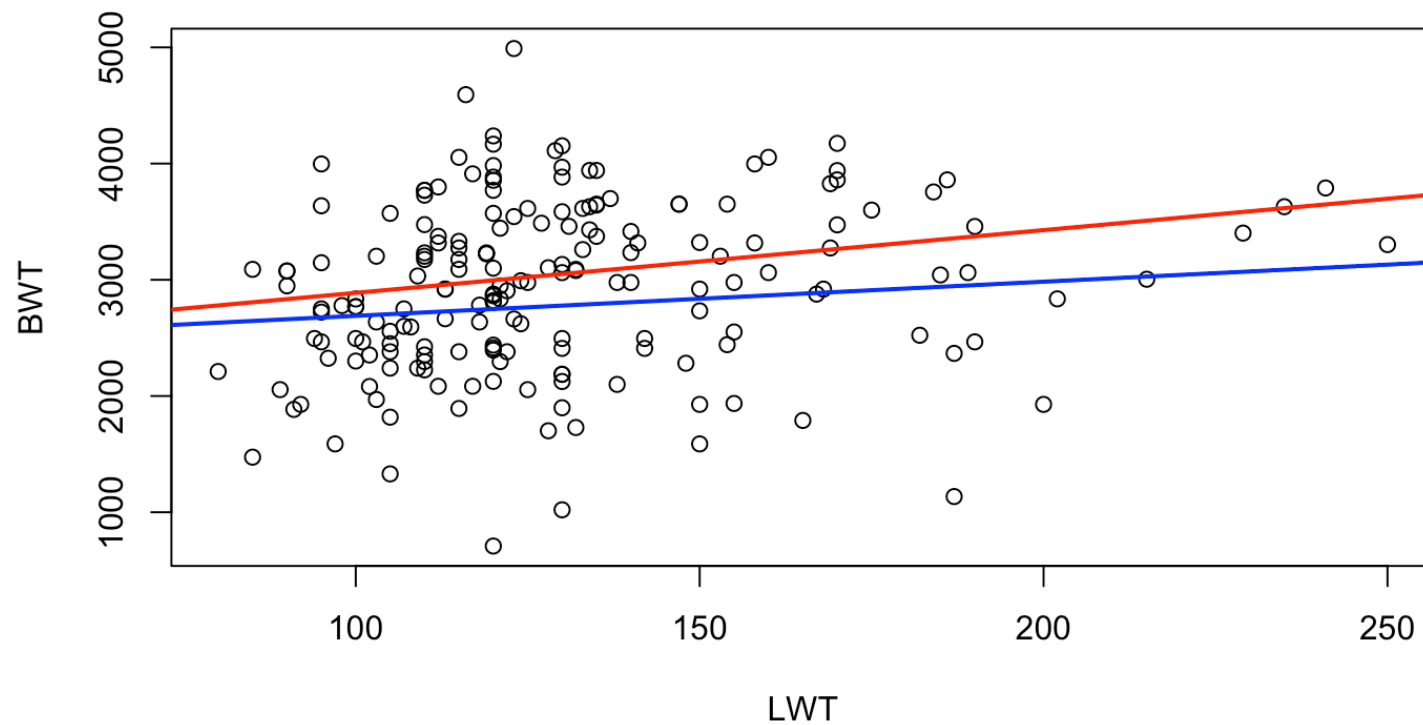
$$y_i = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + \beta_3 x_{1i} \cdot x_{2i} + \epsilon_i$$

with $\epsilon_i \sim N(0, \sigma^2)$.

Lets see how the model is for different values of x_{2i}

Explain coefficients

The estimated model(s)



Model summaries, tests, diagnostics

You should be able to check:

- Are model assumptions met?
- Are there significant effects of LWT and/or SMK?
- How is the model fit in terms of R^2 ? Criticise the model!